



EKF step 2a: Estimator gain matrix (1)

- Recall that the KF gain matrix is $L_k = \Sigma_{\tilde{x}\tilde{z},k}^{-1} \Sigma_{\tilde{z},k}^{-1}$.
- Output prediction error may be approximated:

$$\tilde{z}_k = z_k - \hat{z}_k = h(x_k, u_k, v_k) - h(\hat{x}_k^-, u_k, \bar{v}_k),$$

using Taylor-series expansion around operating point $\{\hat{x}_k^-, u_k, \bar{v}_k\}$:

$$z_k \approx h(\hat{x}_k^-, u_k, \bar{v}_k) + \underbrace{\frac{dh(x_k, u_k, v_k)}{dx_k} \Big|_{x_k=\hat{x}_k^-}}_{\text{Defined as } \hat{C}_k} (x_k - \hat{x}_k^-) + \underbrace{\frac{dh(x_k, u_k, v_k)}{dv_k} \Big|_{v_k=\bar{v}_k}}_{\text{Defined as } \hat{D}_k} (v_k - \bar{v}_k).$$

- Note, much like we saw in Step 1b,

$$\frac{dh(x_k, u_k, v_k)}{dx_k} = \frac{\partial h(x_k, u_k, v_k)}{\partial x_k} \quad \text{and} \quad \frac{dh(x_k, u_k, v_k)}{dv_k} = \frac{\partial h(x_k, u_k, v_k)}{\partial v_k}.$$



EKF step 2a: Estimator gain matrix (2)

- From these results, we can compute:

$$\begin{aligned} \Sigma_{\tilde{z},k} &\approx \hat{C}_k \Sigma_{\tilde{x},k} \hat{C}_k^T + \hat{D}_k \Sigma_{\tilde{v}} \hat{D}_k^T, \\ \Sigma_{\tilde{x}\tilde{z},k} &\approx \mathbb{E}[(\tilde{x}_k^-)(\hat{C}_k \tilde{x}_k^- + \hat{D}_k \tilde{v}_k)^T] \\ &= \Sigma_{\tilde{x},k} \hat{C}_k^T. \end{aligned}$$

- These terms may be combined to get the Kalman gain:

$$L_k = \Sigma_{\tilde{x},k} \hat{C}_k^T [\hat{C}_k \Sigma_{\tilde{x},k} \hat{C}_k^T + \hat{D}_k \Sigma_{\tilde{v}} \hat{D}_k^T]^{-1}.$$



EKF step 2b/c: State/covariance meas. update

EKF step 2b: State estimate measurement update.

- Computes posterior state estimate by updating prediction using estimator gain and innovation $z_k - \hat{z}_k$:

$$\hat{x}_k^+ = \hat{x}_k^- + L_k(z_k - \hat{z}_k).$$

EKF step 2c: Error covariance measurement update.

- Finally, the updated covariance is computed as:

$$\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T.$$



Summary

- EKF makes two fundamental assumptions to generalize KF.
 1. Assume $\mathbb{E}[\text{fn}(x)] \approx \text{fn}(\mathbb{E}[x])$, which is not true in general.
 2. Assume Taylor-series expansion to linearize system equations for covariances.
- Applying these to generic-probabilistic-inference solution yields final three EKF steps.
- EKF algorithm is now derived!